

Multiple choice questions (MCQs):

Choose the correct answer (20 MCQs): Total 10 marks (0.5 mark for each question)

Q1. Why the pulmonary artery blood pressure (BP) is lower than the systemic arterial BP?

- a. The output of the right ventricle is less than that of the left ventricle
- b. The left ventricular wall is thicker than that of the right ventricle
- c. The aorta is more elastic than the pulmonary artery
- ☒ d. The pulmonary resistance is less than the systemic resistance

Q2. Which of the following is true regarding pulmonary circulation?

- ☒ a. It is a low-pressure system that allows for improved gas exchange
- b. It is the larger of the two circulatory systems
- c. The system functions with an increased arterial pressure to circulate through the distal parts of the body
- d. It consists of the left side of the heart, the aorta, and its branches

Q3. Which of the following factors affecting hypoxia brain effects?

- ☒ a. Severity and duration of hypoxia
- b. Presence of cardiac or hepatic diseases
- c. Anemia
- d. Hemiplegia

Q4. A patient presents with a large right pleural effusion. A mass is seen in the apex of the lung and she has cervical lymphadenopathy. A thoracentesis is performed and the following values are observed: • Cell count: 10,000 WBC's, 90% lymphocytes, 5% neutrophils • Pleural fluid LDH (U/L): 600 • Serum LDH (U/L): 200 • Pleural fluid protein (g/L): 4.3 • Serum protein (g/L): 5.5 • Pleural fluid glucose: 50 mg/dl. Which of the following is true?

- a. This transudative effusion is likely caused by heart failure from chemotherapy for lung cancer.
- b. This transudative effusion is likely due to fluid resuscitation provided by treating tumor.
- c. This exudative effusion is likely due to metastatic lung cancer.
- d. This exudative effusion is likely due to tuberculosis.

Q5. Acute respiratory distress syndrome (ARDS) is a rapidly progressive

- a. Non-cardiogenic pulmonary edema
- b. Cardiogenic pulmonary edema
- c. Air way spasm
- d. Pulmonary inflammation

Q6. Patients with idiopathic pulmonary fibrosis typically do not have the following feature?

- a. Exertional dyspnea
- b. Productive coughs
- c. Fine bibasilar inspiratory crackles
- d. An abnormal chest radiograph on presentation

Q7. As a risk reduction strategy, which one of the following vaccination should be recommended to all COPD patients beside a yearly influenza vaccination?

- a. Herpes zoster vaccine
- b. Haemophilus influenza vaccine
- ☒ c. Pneumococcal vaccination
- d. Rubella vaccination

Q8. A 34 year old patient visited the local hospital complaining from dry cough and malaise. his doctor heard a few crackles while performing the general examination on the patient. What type of organism is most likely seen in his case?

- a. S aureus
- b. Mycoplasma pneumonia
- c. E.coli
- d. H. influenza

Q9. Which of the following tests is the standard for the diagnosis of myocarditis?

- a. Echocardiography
- ☒ b. ECG
- c. Endomyocardial biopsy
- d. Electrocardiogram

Q10. Hemoptysis may be found in:

- a. Left ventricular failure
- b. Right ventricular failure
- c. Pulmonary stenosis
- ☒ d. Left to right shunt

Q11. What is the most common cyanotic heart lesion in the neonate?

- a. transposition of the great arteries
- b. total anomalous pulmonary venous return

- c. tricuspid atresia
- d. tetralogy of Fallot

Q12. The most common cause of sudden death during competitive exercise is?

- a. Postoperative congenital heart disease
- ☒ b. Hypertrophic cardiomyopathy
- c. Aortic valve stenosis
- d. Long QT syndrome

Q13. Which of the following is the true statement about Laryngomalacia?

- a. It is deteriorated after age of 6 months
- b. It's the commonest congenital anomalies of larynx
- c. It's is manifested by stridor which worsen in prone position.
- d. It's manifested by stridor which improves by crying

Q14. Arterial blood gas study in cystic fibrosis shows

- a. hypernatraemic hypochloraemic metabolic alkalosis
- b. hyponatraemic hypochloraemic metabolic acidosis
- c. hyponatraemic hyperchloraemic metabolic alkalosis
- d. hyponatraemic hypochloraemic metabolic alkalosis

Q15. How does the act of squatting affect the venous return to the heart?

- a. It increases preload
- b. It decreases preload
- c. It has no effect on preload
- d. It can either increase or decrease preload depending on the individual

Q16. What is the term used to describe the cough that can occur in patients with mitral stenosis?

- a. Ortner's syndrome
- b. Osborne's wave
- c. Osler's nodes
- d. Osler's maneuver

Q17. Which of the following infections is a potential cause of infective endocarditis, which can lead to aortic regurgitation?

- a. Human immunodeficiency virus
- b. Mycobacterium
- ☒ c. Staphylococcus
- d. Salmonella

Q18. Patent ductus arteriosus is a congenital heart defect that is characterized by which of the following?

- a. A hole between the right and left atria
- b. A hole between the right and left ventricles
- c. A persistent connection between the aorta and the pulmonary artery
- d. A persistent connection between the aorta and the inferior vena cava

Q19. What is gross feature of benign hypertension on kidney

- a. Contracted granular kidney
- b. Enlarged kidney with petechial hemorrhage
- c. Posollated kidney
- d. Enlarged Vacuolated kidney

Q20. Which of the following drugs can be used as intravenous therapy for short term treatment of acute heart failure in the hospital setting?

- a. Dobutamine
- b. Hydralazine
- c. Propranolol
- d. Captopril

• Short Answer Question (SAQ):

Write in the following (3 SAQs): Total 3 marks (1 mark for each question)

Q1: List risk factors of pulmonary embolism?

Q2: What are symptoms of Bronchiectasis?

1

2

3

4

Q3: A 75-year-old woman presents in your clinic with complaining that, sometimes when she is going up the stairs she feels faint. Upon taking her history she has stable angina, dyspnoea and heart failure. You suspect she has a heart valve problem.

a) If this woman had mitral stenosis, what would you expect to see on a chest X-ray? (1/2 mark)

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-
-
-

b) What does mitral stenosis sound like on auscultation? (1/2 marks)

On auscultation:

-
-
-
-

•Extended Matched Question (EMQ):

Total 2 marks (One mark for each question)

Q1: Each group of extended matching questions consists of a list of numbered problems (diseases) followed by lettered options (Etiological cause). For each numbered disease select the lettered etiological cause that most closely to the disease. You can use the lettered options once or not at all. Total one mark (0.25 mark for each one)

•Problems:

1. Pulmonary Embolism (B)
2. Bronchiectasis (E)
3. Bronchial Asthma (D)

4. Bronchogenic carcinoma (A)

Options:

- ☒ (A) Chemotherapy
- ☒ (B) Anticoagulant
- (C) Anti-platelet
- ☒ (D) Corticosteroid
- ☒ (E) Antibiotic and drainage
- (F) Anti-fibrotic

Q2: For each clinical scenario, you should choose the most appropriate option from the option list, and you can use the lettered option once, more than once, or not at all (i.e. an option can be correct for more than one clinical scenario and another option can be incorrect for all scenarios). Total one mark (0.5 mark for each scenario)

Options:

- A. Aortic Regurgitation.
- B. Mitral regurgitation.
- C. Acute non-ST segment elevation myocardial infarction (NSTEMI)
- D. Acute ST segment elevation myocardial infarction (STEMI)
- E. Pericarditis.

Scenario (1):

A 45-year-old Egyptian male presented in emergency room (ER) with a crushing chest pain that radiates to the jaw and left shoulder for 30 minutes. He also added history of shortness of breath and nauseous. PO₂ was 96% in room air. His ECG in ER revealed ST-segment elevation in leads V1 till V4 ().

Scenario (2):

A 65-year-old lady presented with shortness of breath on exertion, fatigue, and palpitations. On examination, her jugular venous pressure was elevated, and a pan-systolic, high-pitched "whistling" murmur was heard. The murmur radiated to the left axilla ().

Good Luck